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Problem Set: Slope and equation of the line

1. Find the slope of the line passing through the points A(9,1) and B(4,3)

$m = \frac{y_2 - y_1}{x_2 - x_1}$

Substitution: $m = \frac{3-1}{4-9}$ $m = -\frac{2}{5}$

$m = \frac{2}{-5}$ Final Answer: $m = -\frac{2}{5}$

2. What is the equation of the line in general form that posses through the point (2, -3) with a slope of 5?

$y - y_1 = m(x - x_1)$

Substitution: $y-(-3) = 5(x-2)$

$y + 3 = 5x - 10$ Final Answer: $5x - y - 13 = 0$

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3. Find the equation of the line in general form whose y-intercept is -7 and slope is $-\frac{2}{3}$

$y = mx + b$

Substitution: $y = -\frac{2}{3}x - 7$

$3y = -2x - 21$ Final Answer: $2x + 3y + 21 = 0$

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4. What is the equation of the line in general form whose x-intercept is -2 and y-intercept is 4?

$\frac{x}{a} + \frac{y}{b} = 1$ $a = -2$ and $b = 4$

Substitution: $\frac{x}{-2} + \frac{y}{4} = 1$

$-2y + 4x = -8$ $4x - 2x + 8 = 0$

Final Answer: $4x - 2x + 8 = 0$

5. Find the equation of the line that passes through the points (-1,2) and (-3,-1)

$m = \frac{y_2 - y_1}{x_2 - x_1}$ $\quad m = \frac{-1-2}{-3-(-1)}$ $\quad m = \frac{-3}{-2} = \frac{3}{2}$

Point slope form: $y - y_1 = m(x - x_1)$

$$y - 2 = \frac{3}{2}(x-(-1))$$

$$y - 2 = \frac{3}{2}(x+1)$$

$$2y - 4 = 3x + 3$$

Final Answer: $3x - 2y + 7 = 0$

6. What is the equation of the vertical line passing through the point (-8,0)?

Final Answer: $x = -8$